

The role of Inter-firm dispersion of international marketing capabilities in marketing strategy and business outcomes

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Abstract

Despite their increasing practical and theoretical significance in a globalizing world, marketing capabilities in an international context and their inter-organizational dispersion remain in the shadow of research. Specifically, the usefulness of direct effects of capabilities and their dispersed organization on certain strategies and firm performance is unclear. Drawing on the Dynamic Capability framework, this study investigates boundary conditions that guide the management of capabilities dispersion and its effects on strategic actions and performance. Employing survey data from a sample of firms, findings show that dispersed capabilities are advantageous due to their ability to help firms accomplish key performance objectives indirectly through developing ability to manage overseas elements (adaptation strategy). Furthermore, firms' decisions regarding dispersing capabilities are contingent where market dynamism and coordination moderate the dispersion-adaptation link. The study deepens knowledge on additional ramifications of marketing capabilities, provides new lenses to view marketing dispersion, and offers guidance to managers.

Keywords: Dynamic capabilities; International marketing capabilities; Dispersion; Performance

1. Introduction

Dynamics such as heightened globalization, technological advances, and increasingly networked value chains (Teece, 2014) require firms to access resources not only throughout the organization but also across their boundaries in order to evolve (Hillebrand et al., 2015; Krush et al., 2015). In the marketing domain, activities rooted initially within the firm relentlessly move across its functions and eventually connect with external companies to ensure market-based value creation (Hult & Ketchen, 2017). Indeed, marketing in many companies is moving toward an "extended fabric of partners, marketers and providers" (Day, 2011, p.194) and appears as diasporas of dispersed capabilities that span across external independent organizations (Webster et al., 2005).

The increasing in firms' international involvement, emergence of new foreign operation forms, and changes in marketing's roles, put international marketing (IM) in the limelight as a boundary-spanning and fundamental interface with external environments and organizations (Gnizy & Shoham, 2014; Morgan et al., 2018). Besides, international operation in a globalizing world is the most common strategy that ensures survival and allows firms to achieve higher levels of growth and performance (Spyropoulou et al., 2018). A ten-years-ago McKinsey survey found that dispersion of firms' activities is associated with growth opportunities abroad and in most firms IM activities show more dispersion than other activities. However, with few exceptions, the international perspective of marketing capability organization is disregarded (Schmid et al., 2016; Tan & Sousa, 2015) and there remain important open conceptual and empirical research questions regarding IM capabilities (Morgan et al., 2018). Prior research (Krush et al., 2015) addresses the dispersion of non-IM/domestic marketing capabilities within and beyond firm boundaries but not the inter-organizational dispersion of IM capabilities (i.e., the extent to which capabilities are distributed outside a focal firm boundary and shared out across independent organizations' contributions; henceforth IM capabilities dispersion).

Capability dispersion has deep significance for practice and firms manage their domestic capabilities in ways distinct from those in foreign markets (Vrontis et al., 2009). Notably, relationships between business outcomes and their predictors do not necessarily hold in the firm international context due to the distinctive nature of internationalization (Gnizy et al., 2017). The high levels of complexity and dynamism that stem from the diversity of forces across countries (Leonidou et al., 2013) require firms to orchestrate their capabilities in a more skillful manner. This entails the need for new capacities to ensure that the overall marketing skills and competencies are continuously developed to adapt to changes (Day, 2011; Hillebrand et al., 2015; Morgan et al., 2012; Najafi-Tavani et al., 2016). Thus, international setting is an excellent context for studying such crucial marketing issues.

Further, marketing dispersed capabilities have downstream effects on firms' strategic actions and outcomes (Krush et al., 2015). Firms need to focus on capabilities and strategies to obtain maximum value that drive their behaviors (Gnizy et al., 2014). However, the usefulness of effects of inter-firm dispersed capabilities on certain strategies is unclear (Balabanis et al., 2004; Krush et al., 2015; Pham et al., 2017). One such powerful inevitable strategy for successful internationalization is marketing program adaptation in foreign markets (henceforth adaptation) that requires realization of marketing capabilities (Schmid & Kotulla, 2011; Vrontis et al., 2009). In essence, the IM is the organizational institution that is mostly associated with firms' adaptation.

The above discussed research gaps raise two questions. First, what we know about the value of IM capabilities for organizations and specifically about their inter-organizational dispersion. Second, what is the effect of IM capabilities dispersion on firms' strategic actions and specifically what is the mechanism through which the location of IM capabilities affects adaptation and international performance?

To address these questions, this study adopts a capability-focused approach and uses the Dynamic Capability (DC) framework to examine a model on how IM capabilities dispersion directly and indirectly leads to performance. Dynamic marketing capabilities represent a firm's ability to adjust its marketing activities to cope with changing environments. They reflect the firm responsiveness for creating and delivering customer value propositions in response to market changes. IM capabilities and adaptation activities occur in dynamic environments in which a situational context can establish boundary conditions that determine a variation in relationships between constructs (Spyropoulou et al., 2018; Vrontis et al., 2009). Thus, the model accounts for firms' external (market dynamism; the degree of change or shift in customer needs and preferences) and internal (IM coordination; the integration of related organizational functions) factors, embedded within the DC view, that may moderate the IM capabilities dispersion-adaptation link (Figure 1). By doing so, this study provides new lenses to view marketing dispersion, offers practical guidance, and contributes in several ways.

First, capabilities are a central theme of IM research and their evolving dispersion across organizational boundaries has performance implications and behooves broader empirical evidence (Balabanis et al., 2004; Pitelis & Teece, 2017, 2018). Prior research addresses *non*-IM inter-organizational dispersion (Krush et al., 2015). In an IM context, earlier research examines *intra*- and not inter-organizational dispersion (Gnizy et al., 2017). Furthermore, the research stream on IM capabilities focuses on *non*-dispersed capabilities (Blocker et al., 2011; Chen et al., 2014; Efrat et al., 2018; Morgan et al., 2012; Murray et al., 2011; Spyropoulou et al., 2018). By addressing how IM capabilities assist firms, this study underscores the value of capability dispersion and deepens our knowledge on additional ramifications of marketing capabilities.

Second, existing studies report that weaknesses in firms inside capabilities (e.g.,

architectural; used to formulate and implement strategic decisions) impair the achievement of desired strategic outcomes in international operations (Spyropoulou et al., 2018). To overcome weaknesses, firms may turn to external capabilities, which stresses the importance of studying IM capability dispersion as it may identify an additional mechanism through which capabilities can create value and specify a new path through which internationalization, exercised by IM capabilities and strategies, is linked with performance.

Third, by adopting the DC theory this study extends its understanding not only into an international context but also into dispersed capabilities sphere. Theoretically, the DC view offers insights into IM developments and identifies firms' ability to (re)configure resources as a major element that supports internationalization (Efrat et al., 2018; Liu et al., 2013). Prior research does not employ DC to examine dispersion of capabilities and focuses mainly on firms' existing resources rather than on new upgradeable ones (Balabanis et al., 2004).

Finally, understanding the effect of dispersed capabilities on strategy is of great concern as decisions on adaptation are ongoing and of a central question for IM (Vrontis et al., 2009). Dispersed IM capabilities also present firms with environmental, institutional, and capability-based challenges (Gnizy et al., 2017; Krush et al., 2015). By designing IM capabilities dispersion as a driver of strategy and examining moderators that affect the dispersion-adaptation link, this study offers managers actionable insights and mechanisms that may optimize the use of capabilities to ensure their rational distribution and manage their associated (dis)advantages.

2. Theoretical Background

Marketing capabilities are complex bundles of knowledge, skills, and routines that enable companies to make use of marketing-related resources to carry out marketing tasks and

adaptation to marketplace changes (Day, 1994; Moorman & Day, 2016). The organization of marketing capabilities may be achieved by transforming existing or acquiring new ones from external domains. Specifically, mid and high levels of marketing capabilities (e.g., research, planning, communication & advertising, product specification, pricing) distributed outside the firm have turned an accepted important form to structure capabilities and facilitate competitive advantage and performance (Hillebrand et al., 2015; Teece, 2007; Vorhies & Morgan, 2003).

In the international context, a firm aspiring to internationalize must be able to transfer resources overseas (Balabanis et al., 2004). While numerous studies examine IM capabilities (Blocker et al., 2011; Chen et al., 2014; Efrat et al., 2018; Morgan et al., 2012; Murray et al., 2011; Spyropoulou et al., 2018), they relate to non-dispersed capabilities inside the organization boundaries. Notably, firms' boundary issues are largely capability-related (Teece, 2014). Over the last decades, externalization of organizational functions has grown steadily in significance. Rather than viewing markets as vertical industries, international firms view them more in terms of business ecosystems comprised of peers and providers of complements working together to develop and market superior value propositions to local markets. Capability dispersion has become crucial because cultural and regulatory differences that international businesses face place a greater burden on ecosystems orchestration (Pitelis & Teece, 2017, 2018). Such businesses operate in compound and volatile markets with which they are less familiar and in which they have limited access to overseas resources. Responding to greater complexity requires engagement in interactions with overseas stakeholders who can perform even core activities (Gnizy et al., 2014).

As a reflection of the importance of relationships with third party complementors, many firms provide in their websites, in a dedicated webpage, information about their partners in foreign markets. This information describes how partners contribute to the focal firms' capabilities to exploit advantages in operating successfully in distant markets and afford focal firms local embeddedness and presence on the ground. This reflects the distribution of IM

capabilities among independent organizations. For example, the marketing success of Netflix's streaming services relies on contributions of worldwide third parties such as internet service providers for distribution, production companies for localizing content, and law consultants for supporting legal issues (Hillebrand et al., 2015). Furthermore, national institutions in many countries (e.g., the *Israel Export & International Cooperation Institute*) encourage local companies to connect with cross-border partners' capabilities (e.g., informational, planning) to facilitate and pursue overseas operations.

Blesa and Ripolles (2008) demonstrate that four sets of marketing capabilities are at the core of companies' international decisions that foster their commitment. Outside-in capabilities (e.g., market research and customer relationship management) are used to cope with changes in markets. Inside-out capabilities (e.g., financial management and technology development) are necessary to manage internal resources. Spanning capabilities (e.g., information sharing and coordination mechanisms) are used to integrate the previous two sets of capabilities. Finally, and most relevant to this study, networking capabilities are used to create mutual trust and commitment to share expertise and assets between partners. These sets of capabilities are firm specific because their structure complexity creates barriers to imitation where every firm builds up its own structure relative to rivals to face unique realities of competitive markets, commitments, and future requirements (Krush et al., 2015). Notably, focal firms' ability to integrate their IM activities with third parties in forms such as market and technological ties substitutes for the lack of internal resources, accelerates the flow of specialized information between markets, leads to new competencies, and encompass positive business outcomes. Furthermore, the connection of a firm's resources with those of others alleviates challenges to resources, affects its growth and survival, and determines performance directly and indirectly (Blesa & Ripolles, 2008; Gnizy et al., 2014; Krush et al., 2015; Pham et al., 2017).

This study uses the DC framework to provide an understanding of the IM capabilities dispersion phenomenon. As an extension of the resource-based view with better explanatory

power of the role of dynamic marketing capabilities in operation success, the DC view relates to unique types of capabilities and the utility they provide (Fainshmidt et al., 2016). We distinguish “ordinary” capabilities that support technical fitness, are easily replicable, and reflect “doing things right” from DCs that support evolutionary fitness, are hardly replicable, and have more to do with “doing the right things”. Ordinary capabilities simply allow the management of existing offerings in a static setting and mainly permit growth in non-dynamic low-competition environments with limited globalization (Teece, 2014), which may less fit international setting.

IM capabilities dispersion is a type of inter-firm-cooperation, which according to well-established literature (Pitelis & Teece, 2017, 2018; Liu et al., 2013) can be explained by the DC framework. DCs reflect the firm’s ability to build, (re)configure, and integrate capabilities to maintain competitiveness through combining internal and external assets to cope with new or rapidly changing environments (Teece, 2007). Importantly, DCs reflect firms’ ability to access and incorporate *external capabilities*. As firms integrate various partners’ capabilities for developing theirs, they purposefully extend their resource bases and their capabilities are distributed outside the organization, which form the principles of DCs (Fainshmidt et al., 2016).

Furthermore, DCs relate to the management and strategy domains. The managerial orchestration of capabilities, which is the foundation to the enhancement of processes and exploitation of positions, is fueled by strategy and vice versa. Strong DCs use a (1) sensing, (2) seizing, and (3) transforming (reconfiguring) framework in conjunction with a sound strategy (Teece, 2014). Local partners in overseas markets can uniquely identify and leverage differences and comparative advantages, which mirror the ‘sensing’ and ‘seizing’ functions (Pitelis & Teece, 2017; 2018). The incorporation of partners’ contributions into the focal firm capabilities (e.g., information management, product development) enables focal firms to sense, seize, (re)shape, and (co)create the capabilities within their operation, which in turn create

sources of dynamism in DCs. Alternatively, DCs exhibit features associated with effective processes across firms and can be classified into (1) integration, (2) learning, and (c) reconfiguration (Teece, 2007, 2014). Capabilities dispersion is the capacity to use processes that allow firms to readjust and integrate their resources to match or create market change toward better competitive advantage. Ties with foreign organizations reflect enhanced relational positions that enable focal firms to use third parties' capabilities more effectively to learn and renew resource bases to reconfigure strategy and deal with certain market needs (Liu et al., 2013). Hence, IM capabilities dispersion matches the functional requirements of DC.

3. Model Development

The above discussion on IM capabilities dispersion and DCs highlights additional constructs, all of which relate to the DC approach and are incorporated and integrated in the research model (Figure 1). Firms that possess DCs respond to environmental changes by reconfiguring their resources, which highlight resource deployment and strategic *adaptation* (Teece, 2014). The development of capability-based strategies is crucial and prevail traditional approaches. In this regard, dynamic and renewable capabilities are superior in leveraging firm strategies (e.g., adaptation). DCs also involve composite arrangements of dynamic skills and routines that play an inter-functional *coordinating* role to absorb evolving circumstances to meet the *changing market demands* of foreign customers (Liu et al., 2013; Moorman & Day, 2016; Teece, 2007). Finally, DCs affect *performance* directly and indirectly (Gnizy et al., 2014). In sum, the DCs characteristics highlight their role as agents of evaluation and change that allow companies to appraise adaptations needed to their assets to stay competitive, particularly in ever changing markets (Teece, 2007). The explanatory power of the DC perspective, which has been also used to explore the dynamics of managing ties with external parties and pursue firms' strategies (Gnizy et al., 2014), lies in understanding the conditions

under which internal and external capabilities and market environment enable firms to accomplish their key objectives.

The understanding whether marketing capabilities work through different moderators/mediators is an important research priority because it may direct scholars and practitioners where to look for early performance effects of such capabilities (Moorman & Day, 2016). Particularly, a contingency approach is more realistic to understand adaptation (Hultman et al., 2011; Schmid & Kotulla, 2011; Theodosiou & Leonidou, 2003). Hence, we construct a framework that synthesizes IM capabilities dispersion with contingencies and include theoretically derived contingencies, namely, market dynamism (external firm factor) and IM coordination (internal factor) that may shape the IM capabilities dispersion-adaptation link. In our model, we start with IM capabilities dispersion and examine a pathway that includes adaptation and performance. We specifically establish direct IM capabilities dispersion-adaptation and IM capabilities dispersion-performance links. In the following we elaborate on the model's constructs and its relationships.

IM capabilities dispersion is the focal construct in the model. Scholars identify the dispersion of marketing as being "the configuration of marketing activities and capabilities distributed outside the confines of a centralized marketing department" (Krush et al. 2015, p. 32). Inter-organizational dispersion refers to the extent to which independent partners such as providers, agencies, and consultants contribute to IM capabilities and cause these capabilities to span across the firm boundaries (Krush et al., 2015). The view of IM capabilities dispersion is consistent with the DC approach.

Adaptation refers to the extent to which a firm provides customized marketing solutions in foreign markets. When a firm considers the persistent differences between various country markets and accordingly sets the degree of adaptation activities, a pattern of behavior emerges. The determination of the adaptation level is essential and relates to international strategic decision-making (Schmid & Kotulla, 2011).

The literature discusses the need to balance between adaptation and standardization. Fit-related studies (Hultman et al., 2011; Katsikeas et al., 2006; Zeriti et al., 2014) argue that fit between IM strategy (adaptation, standardization, or a combination between the two) and environmental imperatives enhances performance. However, numerous studies highlight the important role of adaptation and suggest that standardization by itself is not superior for all type of firms. They demonstrate that adaptation is especially valuable for SMEs since their competitive advantage rests on their focus and flexibility in providing customized offerings and marketing programs to meet the requirements of foreign markets (Westjohn & Magnusson, 2017). Gnizy et al. (2014) report that adaptation better satisfies customer requirements and is a key issue for SMEs' engagement, success, and growth in foreign operations. By contrast, MNCs employ worldwide corporate policies and face pressures for cost savings to standardize elements of their IM programs and strategies. In practice, adaptation enables the use of resources and capabilities to achieve best outcomes (Schmid & Kotulla, 2011; Theodosiou & Leonidou, 2003). DCs sharpen a company's global vision toward resource deployment that captures both local adaptation and global integration (Luo, 2000). The ability to adjust dynamically marketing solutions to meet needs of foreign markets challenges the existing capabilities and provides nontraditional advantages. While since the 1980s local differentiation has characterized successful international firms, this success requires DCs (Teece, 2014). Hence, dispersion of IM capabilities relates to adaptation.

Market dynamism refers to the rate of change in customer requirements and preferences (Spyropoulou et al., 2018). In a globalized world firms need to develop customer-focused dynamic capacities (Liu et al., 2013). Specifically, an adaptation strategy involves changing elements of the marketing efforts to fit real or perceived needs of customers in particular markets (Albaum & Tse, 2001; Vrontis et al., 2009). Market dynamism is associated with high degrees of turbulence and changes in international markets (Spyropoulou et al., 2018) in which firms become more dispersed around the globe (Wynarczyk et al., 2013).

Notably, the DC view relates to firms' ability to configure capabilities to cope especially with dynamic environments.

Regarding the internal moderating factor, *IM coordination* reflects the firm's ability to synchronize and integrate related organizational functions to provide maximum effectiveness and efficiency when performing tasks (Leonidou et al., 2013). It refers to the extent to which the functions are interconnected and aligned with IM operations. Where IM capabilities dispersion implies their distribution, many marketing capabilities require the coordination of different organizational units (Gnizy & Shoham, 2014). While intra-firm coordination denotes maximum effectiveness, adaptation is also noted in its relationship to effectiveness (Schmid & Kotulla, 2011). Notably, DCs act as a coordinating mechanism to combine capabilities and resources to accomplish tasks smoothly within the organization (Teece, 2007).

As for the outcome variable of our model, adaptation relates directly to performance and thus is designed as an outcome of IM capabilities dispersion and a driver of performance. Previous studies establish a direct link between adaptation and performance (Theodosiou & Leonidou, 2003). When firms realize their customers' needs and meet them, they create value and are likely to better affect outcomes (Albaum & Tse, 2001; Schmid & Kotulla, 2011). In addition, past research on capabilities focuses on exploring their direct performance effect (Cacciolatti & Lee, 2016; Feng et al., 2017) but not their inter-organizational dispersion effect and not in an IM context. Hence, the model accounts for a direct relationship between IM capabilities dispersion and performance. *Performance* is defined as the outcome of a firm's activities in international markets and is based on market and financial outcomes that reflect the extent of the firm's success in sales and profits. Sales growth and profit indicators at firm level are commonly used to assess performance and reflect the competitive dimensions of international success. Especially,

international operation is a key driver of economic activity and firm growth, and profitability is a priority for marketing managers since it is a primary driver of a firm's stock price (Katsikeas et al., 2000; Lee & Griffith, 2019). Notably, such conceptualization is consistent with extant research that examines the IM capabilities-performance (Blesa & Ripolles, 2008; Pham et al., 2017) and adaptation-performance (Schmid & Kotulla, 2011; Westjohn & Magnusson, 2017) relationships. Specifically, it is in line with prior research that studies IM capabilities dispersion (Gnizy et al., 2017). This well-defined conceptualization also represents a dominant aspect in research to view the performance outcomes of marketing (Katsikeas et al., 2016).

4. Hypotheses Development

4.1. IM capabilities dispersion and adaptation

Firms' success rests on the development and deployment of assets and building relationship capital. International firms need dynamic connectivity capabilities to provide value-added propositions to overseas customers (Cano-Kollmann et al., 2016). This requires companies to promote responsiveness and flexibility achieved through DCs (Teece, 2014). DCs in international environments imply building localized capability combinations to serve markets and guide firms to create and deploy situation-specific knowledge (Liu et al., 2013). Increased knowledge on markets motivates firms to develop offerings that better fit the local needs (Albaum & Tse, 2001).

To overcome weaknesses encountered in foreign markets and to develop and deliver solutions in accordance with customer specific needs, firms can engage with partners (Liu et al., 2013; Teece, 2014). By dispersing capabilities and utilizing third parties' more-reliable and specialized knowledge on foreign markets, focal firms gain access to broader and more accurate market perspectives and understanding than they

would have on their own. Consequently, firms can promote responsiveness to customer needs and utilize flexibility by considering alternatives that partners can offer. The incorporation of partners' contributions into focal firms' capabilities reflects openness to new ideas that alter organizational practices and act as a valuable resource that facilitates aggressive adaptation (Calantone et al., 2006). Furthermore, the reliance on partners' capabilities may equip firms with more confidence in adapting offerings to the differences identified in foreign markets. These support firms to stimulate an adaptation strategy that relates to the challenge of determining the customization level of marketing strategies.

Finally, firms need not only find the optimal level of adaptation of their marketing-mix; but also decide upon their proper organization that assists in sustaining competitive advantage. High degrees of decentralization regarding the implementation of IM activities are more favorable to local adaptations (Schmid et al., 2016). Notably, capabilities dispersion reflects a decentralized structure that affects adaptation levels (Schmid & Kotulla, 2011). In addition, firms that aspire to adapt their foreign marketing-mix activities need to have high international competences. Such firms can take full advantages by incorporating external capabilities to build their own competencies (Hillebrand et al., 2015). By increasing connections with overseas markets through capability partnership, firms can more effectively build capabilities to meet local conditions (Liu et al., 2013).

In sum, by dispersing IM capabilities, firms are associated with better ability to manage scattered capabilities in various heterogeneous contexts, which may place them in a better position to adjust their offerings and respond to different and specific needs in foreign markets. Solely relying on self-capabilities to promote adaptation may signal a failure. Thus, we hypothesize that:

H₁: Inter-organizational IM capability dispersion is positively related to marketing program adaptation.

4.2. Moderating effects of the IM capabilities dispersion-adaptation link

Dynamic international environments are associated with increasing variations in customers' behaviors and requirements, which pose challenges to strategy accomplishment (Spyropoulou et al., 2018). Specifically, market dynamism is a factor that affects the pressure on firms to adapt their offerings (Calantone et al., 2006; Lee & Griffith, 2019). Under frequent and unpredictable changes in foreign market demands, firms face difficulties to follow, evaluate, and understand customer preferences and needs (Blocker et al., 2011), which may impair their ability to forecast accurately and predict appropriate marketing actions. Such variations pull firms towards adoption of additional resources to respond satisfactorily to customers' changes (Jaworski & Kohli, 1993; Vrontis et al., 2009). The DC view suggests that firms reconfigure their capabilities to adapt marketing strategy to keep pace with shifts as DCs allow meeting customers' changing demands and reacting accordingly (Liu et al., 2013; Tan & Sousa, 2015). These necessitate access to external resources and encourage firms turning to outside capabilities (Teece, 2007). The integration of localized up to date capabilities that emanate from external partners' contributions becomes an important management instrument and imperative to support adaptation efforts (Wynarczyk et al., 2013) and bring about new synergies to help renewing capabilities (Teece, 2007). If firms fail to do so, they may fall to the inertia trap that limits their ability to adapt to environmental changes and harms their efforts to respond to changed customer's preferences.

In sum, when environments become more customer turbulent, the consensus-based decentralized capabilities inferred by dispersion may increase adaptation. Thus, we hypothesize that:

H₂: The relationship between inter-organizational IM capability dispersion and marketing program adaptation is positively moderated by market dynamism.

Capabilities are in nature a coordinating mechanism that binds different resources to accomplish activities properly (Leonidou et al., 2013). Specifically, marketing capabilities are collective routines that serve as a critical coordinative means (Krush et al., 2015). Cross-functional coordination is imperative for better assembling the right mixture of organizational resources to create IM strategies (Leonidou et al., 2013). Intra-firm coordination within the IM dispersion literature is considered a crucial element to leverage firm resources and essential in the design and execution of IM adaptation strategies (Gnizy et al., 2017; Gnizy & Shoham, 2014).

IM operations and specifically adaptation implementation cut across functional boundaries rather than be the sole responsibility of marketers. Since adaptation requires continual reconfiguration of resources to meet enduring changes, intra-firm interfaces become critical. When in addition multiple specialists such as external parties are involved in IM operations through capabilities dispersion, coordination becomes further imperative (Leonidou et al., 2013). Firms that operate in the form of increased dispersion experience high complexity that emanate from differences across countries and especially from the flow of assets throughout the organization's internal and external structures. Such firms that transfer resources among different partners need to maintain effectual communication mechanisms in the form of increased coordination to effectively adapt strategies to respond to different environments (Liu et al., 2013). Therefore, when the level of IM coordination increases, the impact of dispersed capabilities on adaptation becomes stronger.

Finally, coordination mechanisms better integrate internal and outsourced activities and resources, allow cross-fertilization among them, and facilitate the use of overall capabilities. They help achieving common understanding of IM decisions and actions among various collaborates regarding required customizations. Higher coordination enables coherence beyond the firm's boundaries to deal with drawbacks (e.g., consistency)

that IM capabilities dispersion may carry en-route adaptation. Thus, we hypothesize that:

H₃: The relationship between inter-organizational IM capability dispersion and marketing program adaptation is positively moderated by IM coordination.

4.3. Effects on international performance

Internationalization requires firms to adapt local circumstances in host countries. Theory- and practical-based evidences support the positive relationship of adaptation related activities with successful internationalization (Liu et al., 2013; Zeriti et al., 2014). The adaptation process helps define firms' competitive advantage, which in turn affects their performance (Albaum & Tse, 2001). While decisions on adaptation are cost-relevant and relate to financial performance, most multinationals believe that adaptation strategy facilitates sales growth and profitability (Calantone et al., 2006).

With adaptation, firms gain advantages in better communicating with foreign markets. Through effective engagement with customers, firms can offer value-added offerings, which in turn reap financial returns (Lee & Griffith, 2019). Specifically, greater adaptation enables firms to more effectively deliver benefits to international customers because it ensures their satisfaction and paves the way for increased sales and profits (Gnizy et al., 2014). In addition, firms that employ adaptation strategy are likely to enjoy further advantages such as negotiation and accommodation to local regulations that enable them to outperform others (Calantone et al., 2006). Hence, adaptation enables profitability through higher sales accrued from a better exploitation of differences across countries and thus particularly improves foreign sales and profits (Schmid & Kotulla, 2011; Theodosiou & Leonidou, 2003). These arguments are consistent with the broader literature that postulates and confirms positive relationship between IM program adaption strategy and performance, especially in SMEs (Calantone et al., 2006; Gnizy et al., 2014; Westjohn & Magnusson, 2017).

In sum, firms' actions that tailor the IM program to meet specific needs of target markets benefit performance. Thus, we hypothesize that:

H4: Marketing program adaptation is positively related to international performance.

International capabilities directly and positively affect firms' international sales and profits (Blesa & Ripolles, 2008). For example, export-marketing capabilities such as distribution, promotion, and pricing enhance export financial performance (Pham et al., 2017). Notably, the dispersion and heterogeneity of marketing capabilities that result from connecting to other companies' resources reflect their dynamism and have direct implications for performance and specifically meet challenges of profit expectations (Krush et al., 2015; Moorman & Day, 2016). While a firm's dependence on exporting activities is a source of revenues and profits, the incorporation of partners' contributions into the focal firm capabilities reflects its openness to marketing innovation, which is a critical factor in improving international performance (Calantone et al., 2006). Flexible capability alternatives such as new product development and efficient distribution systems that third parties can offer bring to better performance growth for the focal firm (Katsikeas et al., 2000). Such external resources facilitate the firms' decision process, reduce cycle times of activities, and improve learning (Hillebrand et al., 2015), which can all be translated into higher performance (Pitelis & Teece, 2017, 2018) and specifically into higher sales growth (Feng et al., 2017). In addition, companies that efficiently gain access to local markets through partners compensate for deficiency in internal resources, decrease associated costs, facilitate the firms' operation abroad, and lead to better business outcomes (Gnizy et al., 2014; Krush et al., 2015; Liu et al., 2013). Thus, we hypothesize that:

H5: Inter-organizational IM capability dispersion is positively related to

international performance.

5. Methodology

5.1. Research context

The study's research context is Israel, which is an important empirical setting because of its substantial reliance on international operations as its economy engine. Export revenues accounted approximately for 40% of the country's GDP in 2017 and Israeli firms have been showing a constant growth over recent years in international operations (*Israel Export & International Cooperation Institute 2017; IEICI*). Israeli firms are geographically distant from their foreign markets and customers, and thus the propensity to incorporate third parties' resources to build and orchestrate overall capabilities is particularly relevant. Notably, due to the national government's strong emphasis on promoting export and international engagement, governmental institutions encourage Israeli firms to facilitate and foster cooperative market relationships with overseas partners to connect with their resources when they pursue international operations (*IEICI*). Indeed, Israeli companies connect with third parties to achieve competitive advantage, growth, and success in foreign markets (Friedrich et al., 2014). Thus, our study on IM capabilities dispersion in this setting is timely. Finally, by focusing on Israel, the study has also the potential to shed light on firms' conduct in similar other economies of the world that share similar institutional settings. Israel is viewed as a developed country tending to have a Western orientation. Prior research supports the notion that marketing capabilities appear more prominent for firms that operate in developed rather than other countries (Pham et al., 2017).

5.2. Data collection

Data were collected from international Israeli firms through a professional panel company that conducted an online survey. The market research company approached firms

and key informants (typically mid-/high-level IM managers) according to a pre-defined profile that fits this study's context. Only firms with at least twenty-five employees (e.g., to more precisely capture constructs such as coordination) were approached. Startup companies were not eligible to participate since their business model or performance outcomes may be unstable. In addition, firms that had been in business for less than 3 years were excluded, as they may not have a stable international business setting yet. The design aimed to achieve wide but relevant distributions for respondents' position and seniority (full-time managers at mid-level and higher), experience (with the current firm), and knowledge of the subject matter.

To guarantee a high level of participants' appropriateness and reduce potential bias at the key informants and firms' levels, the questionnaire included two 7-point items that assessed respondents' familiarity with and confidence in addressing issues covered in this research. Respondents with low-level scores on these items (below 4 out of 7) were excluded. The items averaged 4.92 for familiarity and 5.26 for confidence suggesting high levels for these measures. These steps ensured respondents were well positioned and qualified to provide insights about the study's subject matters and items. The respondents' characteristics described below support this conclusion. The questionnaire started with a brief explanation and guidelines on the research background and objectives. Participants were told that the items related to firms' IM capabilities and relationships with overseas partners. Out of 4,568 participants who were approached by the managing company, 487 were deemed qualified to complete the survey. After excluding respondents/firms according to their appropriateness described above, the data collection process resulted in 192 completed and usable responses for an effective response rate of 39.4%, which is in accordance with similar IM capabilities investigations (e.g., Pham et al., 2017).

A series of t-tests that compared early and late respondents with regard to the

study's constructs found no significant differences at the 5% level and thus ensured that delay bias was not likely to be a problem (Armstrong & Overton, 1977). We also compared the differences between participants and non-participants with regard to key characteristics and obtained again non-significant statistical differences providing further evidence for lack of selection/non-response bias.

5.3. Sample description

Our sample reflected a cross-section of executive rank, function, and experience. 43.2% of the respondents were senior/top management executives (e.g., CEOs, VPs, division managers) and 56.8% were senior/mid-level managers (e.g., directors, group/product/brand leaders) with positions in domains such as general management, marketing, sales, finance, R&D, and operations. 51.8% exceeded 10 years of personal management experience. Participants averaged 8.2 years in their current firms.

The firms in our sample operated mainly in B2B markets (the 7-point B2B-B2C focus scale indicated an average score of 2.3). The sample represented a cross-section of industries such as hi-tech (37% of the firms), retail and consumer goods (24%), medical/health (11%), defense/security (8%), aggrotech (7%), aviation/transportation (4%), and rest in diverse industries (9%). This multi-industry design allows greater variability, reduces the likelihood of sampling bias, and enables greater generalizability (Morgan et al. 2012). Additionally, the firms in the sample ranged over various characteristics. They represented mainly SMEs (24.5% were large [with >250 employees], 67.2% medium [<250], and 8.3% small [<50]; Wright's [2010] classification). Notably, SMEs usually rely on external partners' capabilities in foreign markets (Gnizy et al., 2014). The firm average number of employees was 466 and average firms' age was 38 years. More than 65% of the firms had been in business for more than 20 years and only less than 4% had been in business for less than 5 but more than 3 years. The firms had been

involved internationally for an average of considerable 28 years. Notably, the firms' characteristic diversity ensures data richness and increases observed variance.

We asked respondents to indicate their firm's international scope (i.e., the geographical regions in which the firm mainly operates). The distribution of regions, namely, mainly to North America (36.9% of the firms); then to Europe (34.8%); Asia 13.6%; and other countries (14.6%) was consistent with national data on international geographical spread (*IEICI*). Hence, the sample represented a broad cross-section of the international activities of Israeli firms.

5.4. Measurement scales

All measures (Appendix), with minor adaptations to the study's context, were based on existing scales in the literature. All items were translated for use with Israeli respondents and were back translated to achieve linguistic equivalency. For face and content validity, two marketing academics and three executives commented on the appropriateness, relevance, clarity, wording, and conciseness of the items. In addition, a preliminary pretest of the questionnaire was conducted with twenty managers, prior to the main data collection stage, which confirmed the measures' content validity and their initial statistical reliability and validity.

The scale for *IM capabilities dispersion* was adopted from Krush et al. (2015). We used their inter-organizational marketing capability dispersion items adapted to the study's context. It measured the extent to which capabilities span across organizations. Respondents were asked to determine the extent to which independent partners contribute to IM capabilities. Notably, marketing capabilities are more than just the marketing-mix capabilities and the DC literature focuses on the importance of mid- and higher-level marketing activities (Najafi-Tavani et al., 2016). Therefore, the scale incorporated most commonly key marketing capabilities that are critical to achieving organizational goals in value creation (Srivastava et al., 1999), which are also congruent with primary IM capabilities (Balabanis et al., 2004).

While previous studies focus on individual capability (Feng et al., 2017) and do not investigate what type of marketing capabilities should be used, in practice, distinct capabilities coexist. Thus, the measure of dispersion suggested multiple capabilities. Such operationalization also addresses potential sources of endogeneity (e.g., measurement issues; Boehe & Jimenez, 2016). Importantly, the business processes supported by IM capabilities dispersion are operationalized based on Teece's (2007) classification of DCs: integration (operation improvement/effectiveness such as channel management capability), learning (sourcing analysis and forecasting such as planning capability), and reconfiguration (network design and optimization).

Marketing program adaptation was adopted from Gnizy et al. (2014) and was designed to capture a range of marketing related capabilities that are associated with the elements of marketing activities. Importantly, our conceptualization of adaptation went beyond the basic operationalization of the four P's marketing mix. The scale included additional factors such as downstream/upstream supply chain management and sales activities to make it widely congruent with the activities of IM capabilities dispersion.

For the moderators, we used an adaptation of Spyropoulou et al.'s (2018) *market dynamism* scale to capture the degree of changes in customers' needs and emergence of new offering in overseas markets. Adapted Cadogan's et al. (2012) export coordination scale was used to measure *IM coordination*.

As for the terminal dependent variable, the literature suggests varying operationalizations of international performance. In the past, the use of a single item approach to measure performance has been criticized and the multi-indicator approach has been supported. However, the decision on the best approach remains inconclusive and numerous scholars encourage and employ the single item approach (Calantone et al., 2006; Katsikeas et al., 2000; 2016). Hence, we captured two dimensions of *performance* separately - sales (indicator of market performance) and profits (indicator of financial

performance). The item questions included frame of reference. These two clean indicators reflect different aspects of performance and are likely to be affected differently by the study's independent variables. For instance, a certain strategy might help firms achieving higher sales, but this may incur additional costs that reduce profit potential. Notably, these indicators are practical, likely to have the same meaning across firms, and are widely used in research on the performance outcomes of marketing (Katsikeas et al., 2000; 2016).

Finally, based on prior empirical studies (Gnizy et al., 2014; Pham et al., 2017), we included control variables to account for extraneous sources of variation in the dependent variables (Appendix).

5.5. Reliability and validity

All scales' items exhibited acceptable psychometric properties that ensured their validity and reliability. In addition, convergent and discriminant validity tests at both the item and construct levels were performed (Voorhees et al., 2016).

First, we used previously validated and reliable scales. Second, to verify the scales structure, an exploratory factor analysis (EFA) was conducted and revealed that the measures were unidimensional where all items loaded onto their respective factors with high item loadings ($\geq .67$, Hair et al. 2010). In addition, a confirmatory factor analysis (CFA) on all constructs yielded acceptable fit statistics ($\chi^2=1439.5$, $df=992$, $\chi^2/df=1.45$, $p \leq .01$, RMSEA=.06, IFI=.91, CFI=.98). Third, all scales exceeded recommended threshold of .70 for alpha and .60 for composite reliability (Table 1) suggesting acceptable degrees of internal consistency for each construct. Fourth, constructs in a research model should have relatively low inter-correlations (i.e., $< .6$, Voorhees et al. 2016). The model's constructs were not highly correlated ($< .43$, Table 1). Fifth, in support of scales' convergent validity, all model scales' average variance extracted (AVE) estimates exceeded .50 (ranging .57-.61). Finally, in support of discriminant validity, the square root

of each construct's AVE was greater than the construct's correlation with other constructs (Table 1). The last test is mostly recommended for studies in marketing to support discriminant validity. Furthermore, the model analysis results reported below show that each construct in the model affected or was affected differently. Hence, our statistically significant variables were supported by the data, were distinct and not just an artifact of modeling or empirical reflections of each other, in further support of discriminant validity (Voorhees et al., 2016).

[Insert Table 1 about here]

5.6. Common method variance (CMV) assessment

We followed ex-ante and ex-post procedures recommended by Podsakoff et al. (2003) to minimize CMV concerns. *Ex-ante* included the following research design-related strategies. First, we applied a systematic design of the questionnaire and development of the measures (e.g., division into sections, multiple scale lengths and formats) to ensure clarity, etc. Second, we told the respondents that the questions had no right/wrong answers and thus they should answer honestly to reflect their insights. They were also guaranteed with anonymity and confidentiality. Third, respondents could not move forward along the survey before completing answers to all current displayed questions. Finally, our model entailed complex relationships thus it would have been unlikely for respondents to anticipate the relationships under investigation.

Ex-post strategies included the following (Podsakoff et al., 2003). First, we conducted a single-factor test. In an EFA that included all indicators of the study's constructs, the first factor accounted for only 36.8% of the variance, suggesting no single factor is responsible for most of the variance in the measures. Further, a CFA in which all items were modeled as indicators for a single factor yielded a poor model fit (chi-square =402.18, $p < .01$, d.f.=76, RMSEA=.14, NNFI=.47, CFI=.49) suggesting that a single

method factor was not a viable solution. Second, we used the ex-ante and ex-post marker variable approach. The ex-ante procedure requires the a-priori use of a study variable that should theoretically not relate to the model's latent variables (Lindell & Whitney, 2001). The correlations between the item (i.e., respondents' readership of a marketing journal) and the model's constructs were unrelated. Furthermore, we adopted ex-post identification of a marker variable by using the second-smallest correlation among the study's constructs ($r=.03$, $p>.05$). We calculated the CMV-adjusted correlations (Malhotra et al., 2006) and then compared them to the original unadjusted correlations. The correlations remained stable and maintained their statistical (in)significance. Finally, we tested for multicollinearity by variance inflation factors (VIFs), which were all well below the recommended value of 10 (≤ 1.4 ; Hair et al., 2010). Collectively, these tests reinforce the lack of CMV evidence.

6. Analyses and Findings

6.1. Descriptive results

Since it is the first study that examines inter-organizational dispersion of IM capabilities, it is worth discussing descriptive results for the dispersion construct. The mean of the overall capability dispersion measure was 4.55 (out of 7) suggesting that firms distribute their IM capabilities and activities among external independent organizations. In other words, firms conduct certain IM activities by outsourcing them to span across organizational borders. This is consistent with the view that firms' marketing activities are increasingly nested and carried out today within networks of partners (Pitelis & Teece, 2017, 2018). Further examination of the single items of the IM capabilities dispersion measure reveals that their means ranged between 3.1 and 5.1. These scores indicate several insights. First, the various dispersed capabilities have relatively high scores, which again communicate the contributions of third parties to firms' IM operations. Second, there are

no capabilities that are maximally or minimally dispersed. While IM execution in these firms involves external parties, no IM task is entirely carried out of the firm. For example, certain capabilities such as information and communication management are depicted relatively with high dispersion (mean of 5.1 and 4.7, respectively) while others such as pricing (mean of 3.1) are indicated relatively with low dispersion. Finally, it may be possible that some dispersed capabilities are more crucial than others are for the development of advantage.

6.2. Hypotheses testing

Hierarchical (moderated) regression analyses were employed to test the hypotheses (Hair et al., 2010). This method better fits this study's design and setting (e.g., sample size, numerous controls). Notably, recent research on international capabilities (Pham et al., 2017) used this method. In addition, prior research (Gnizy et al., 2014) that applied a DC-strategy-performance path model used regression analyses and conducted supportive SEM path analyses that corroborated the main regressions' findings. In the analyses, the first model included only the controls, followed by addition of the independent variables to the second model. To test for moderating effects, a third model with the relevant interaction terms (in addition to the variables in the second model) was run. Further steps ensured the regression models met established guidelines (Hair et al., 2010). First, the sample size met guidelines for a minimum ratio of observations/independent variables. Second, in hierarchical regressions, multicollinearity can be a concern and affect the results. However, all model's construct fell within a moderate to low range of inter-correlations ($<.43$; Table 1). In addition, the VIF scores for all independent variables in the models were below 1.4, well below the acceptable cutoff of 10. Thus, multicollinearity did not affect our estimations. Third, products of mean-centered variables involved in the interaction terms were used to test the moderation effects and to reduce multicollinearity. Finally, all models

were significant ($p \leq .01$) with high levels of explained variances. The increase in R^2 in all models (compared with the control variable model and importantly, with the model without the interaction terms) indicated that the independent variables explained supplementary variance. Notably, the adjusted R^2 values revealed no significant loss in predictive power compared to the R^2 values indicating results are generalizable (Hair et al., 2010).

Table 2/Models 1-3 report the standardized regression coefficients for the effects on adaptation. The effect of IM capabilities dispersion on adaptation was positive and significant ($\beta = .143$, $p \leq .05$), in support of H_1 . Such evidence has not been shown in prior research. While prior research demonstrates the positive link between marketing implementation capabilities and export performance (e.g., Pham et al., 2017), it has not accounted for capabilities dispersion. H_2 tested the positive effect of market dynamism on the IM capabilities dispersion-adaptation link, which was significant ($\beta = .601$, $p \leq .01$), in support of H_2 . Notably, the relatively large magnitude of the coefficient for this interaction indicates the important role that market dynamism plays in the dispersion-adaptation relationship. Similarly, H_3 tested the positive effect of IM coordination on the IM capabilities dispersion-adaptation link, which was significant ($\beta = .134$, $p \leq .05$), in support of H_3 . This result reflects the necessity of cross-functional coordination for dispersed capabilities because international firms are confronted with diverse, complex, and volatile environments. As for the controls, only the B2B/B2C focus was related to adaptation in the complete Model 3. Finally, results also show that the significant addition of the interaction terms to the main effects model increased R^2 by 17% ($p \leq .0$; Model 3), suggesting that the interaction model provides the best overall model fit for predicting adaptation.

[Insert Table 2 about here]

As for the coefficients of the effects on international sales and profits

performances, in the complete Models 6 and 9 that included IM capabilities dispersion and adaptation, only adaptation positively related to sales ($\beta=.137$, $p\leq.01$) and profits ($\beta=.099$, $p\leq.01$), in support of H4. These results are consistent with prior findings in the literature that adaptation offers a means to differentiate and better serve specific foreign market segments and thus enhance performance (Calantone et al., 2006; Westjohn & Magnusson, 2017). Notably, in these models IM capabilities dispersion did not relate to performance ($p>.05$). Hence, H5 is not supported. These finding must be interpreted considering additional analyses discussed below. As for the controls, we included dynamismXadaptation and coordinationXadaptation interaction terms as control predictors of performance since as discussed above dynamism and coordination can make adaptation more effective. In addition, prior research identifies the environment as a moderator of the adaptation-performance link and specifically market dynamism may make adaptation more necessary to attain higher performance (Schmid & Kotulla, 2011). These two interactions were marginally related to sales and profits performances (Models 6, 9).

6.3. Additional and robustness checks

Additional analyses were conducted to shed light on our main findings. We conducted a deeper examination of the unsupported direct effect of dispersion on performance (H5) and tested whether the IM capabilities dispersion-performance relationship is mediated by adaptation. As suggested by Baron & Kenny (1986) and confirmed by Preacher & Hayes (2008), we tested for full mediation that should predict (1) a positive dispersion-adaptation relationship, which is shown in Table 2/Model 3, (2) a positive dispersion-performance relationship, which is shown in Models 5+8, (3) a positive adaptation-performance relationship, which is shown in Models 6+9, and (4) a non-significant dispersion-performance relationship after the effect of adaptation is controlled, which is shown in Models 6+9. In this latter condition, the effect of IM capabilities

dispersion on performance became statistically insignificant after adaptation, which was statistically significant, was added to the analysis. In full mediation, the predictor construct loses its power to affect the criterion construct except through a mediator (Hair et al., 2010). Hence, our findings suggest a mediated relationship of IM capabilities dispersion with performance through adaptation, which highlights the important role of adaptation in implementing dispersion. In other words, firm performance (sales, profits) may not be influenced by dispersion unless materialized through adaptation. Notably, DCs affect performance indirectly, act as the glue that binds various organizational resources to enable their deployment for maximum advantage, and lead to superior performance (Gnizy et al., 2014).

Further checks were conducted to test the consistency and robustness of our model and to assess the stability of our findings. Notably, inconsistency is the main threat to endogeneity (Antonakis et al., 2014). First, since our model includes multiple controls, we ran the main models without the controls to see if our basic hypotheses are grounded statistically and if similar results are obtained. Results indicated that the significance levels of the coefficients' signs remained unchanged. Thus, our hypotheses are statistically grounded. Second, additional steps were taken to deal with reverse dynamics that may take place in our model. For example, while our study posits the impact of IM capabilities dispersion on adaptation, it is also possible that the level of adaptation influences firms' decisions to disperse capabilities. Similarly, we posit the impact of adaptation on performance. However, the more profitable firms are, they can opt for higher levels of adaptation. To address potential reverse effects, we ran a series of alternate models with reverse links. Notably, all alternate models underperformed, explained less of the variance sometimes indicating significant decrease, and had a weaker fit than the primary model. Moreover, some of their effects were insignificant. Thus, reverse dynamics did not seem to

be an issue and our original model best explains the relationships between the constructs. Third, CMV, another cause for inconsistent estimates that is related to measurement error (Antonakis et al., 2014) has been shown above to unlikely be a concern in this study. Finally, as noted above, the operationalization of IM capabilities dispersion also addresses potential sources of endogeneity. Additionally, firms' decisions regarding dispersing capabilities are complex and are relative rather than absolute or random. By examining moderators that may change the impact of IM capabilities dispersion on adaptation, we inherently address issues related to endogeneity (Murray et al., 2012).

7. Discussion

This study investigates boundary conditions that guide the management of inter-firm dispersion of capabilities and its downstream implications on strategic actions and performance. While international firms are increasingly dispersing their capabilities across their boundaries, the certain utility and consequences of dispersing IM capabilities and the mechanism through which they are realized are under-researched. Notably, the study supplements knowledge on the dispersion of IM in an intra-firm context. IM capabilities dispersion reflects the more novelty and complexity in the practices and strategies of modern international firms. It provides a theoretical vein to expand on a vital aspect of business ecosystems that are made up of worldwide organizations working together to develop and sustain markets and value propositions (Pitelis & Teece, 2017, 2018). The dispersion highlights dynamics and capability augmentation and emphasizes the better understanding of the nature and source of capabilities that lead to improved performance.

This study includes adaptation as a pathway between IM capabilities dispersion and performance. The support from this study for the value of integrating a strategic action into *dispersed* capabilities through the capability-strategy-performance framework provides further evidence of the usefulness of applying DCs. The focus on IM capabilities

dispersion as a DC enriches the understanding of the DC framework into international and dispersion contexts. This should pave way for researchers to investigate DC dispersion as a driver of other aspects of international strategy. Furthermore, our findings broaden the strategic management research by showing that ongoing engagement with ecosystem partners contribute to the asset augmentation, exploitation, extension, and renewal of focal firms. This may lead to the migration of the locus of value creation from the focal firm to the level of the business ecosystem (Teece, 2014).

Our results differ from the ones of Krush et al. (2015) who studied dispersion of non-IM marketing capabilities and found they affect firms' institutional factors (e.g., marketing's influence) and outcomes. While studies on the capabilities of firms that operate in the global marketplace lag those in domestic market settings, our study provides a key understanding of the organization of IM capabilities and shows that their dispersion enhances adaptation and has also indirect effect on performance through adaptation. In addition, firms' external (market dynamism) and internal (IM coordination) moderators explain the success of dispersion in enhancing adaptation. While the distribution of IM capabilities to external partners emerges as a driver of adaptation, firms can use inter-firm capabilities to establish a baseline to determine the level of adaptation because dispersion through strong DCs help organizations stay relevant to marketplace needs. By dispersing capabilities, the focal firm alters its resource base and is no longer restricted to its capability boundaries. It can gain access to a broader set of resources than it would have on its own. Dispersion allows for value-added benefits through the external partners' specialization that enables gaining better understanding of enduring changes in globalizing marketplaces and increases the need to dynamically (re)develop new capabilities. This determines the focal firm's ability to build, reconfigure, and integrate internal and external resources to address ever-changing business environments, as the DC approach suggests.

The study also exhibits the pertinent role that adaptation plays in leveraging outcome advantages of SMEs since it surfaced as a predictor of performance that further support earlier works (Calantone et al., 2006; Westjohn & Magnusson, 2017). Importantly, adaptation also emerged as having a significant mediation effect. Appropriate investment in adaptation facilitates the contributions of third parties for leveraging sales and profits. Put it differently, for dispersed IM capabilities to function effectively, they should be applied through firm strategies (e.g., adaptation). These offer evidence about the complex relationship between firms' marketing capabilities and performance, provide empirical support for the argument that DCs and business strategies codetermine performance (Teece, 2014), and corroborate prior research (Feng et al., 2017). Evidence on a mediating role of adaptation has not been shown in prior capability research that rather demonstrated how marketing capabilities mediate other firm strategies (e.g., market orientation) in their effect on performance (Pham et al., 2017; Spyropoulou et al., 2018). Thus, theoretically this study suggests new insights on the significant role of adaptation. It also adds value to the literature on drivers of adaptation that suggests only a small subset of empirically examined antecedents (Schmid & Kotulla, 2011) by proposing capabilities dispersion as such a driver.

Our findings support the notion that dynamic changes in markets positively moderate the relationship between capabilities dispersion and adaptation. Thus, when the perceived degree of shifts in customers' needs is higher, a strategic option for firms could be to turn to outside capabilities. While higher levels of market dynamism may make the management of capabilities dispersion more complex, higher degrees of third parties' contributions provide access to greater customer-based information and resources that can assist in fast and flexible response for customer needs.

In line with the DC perspective that views DCs as a firm's ability to strategically

integrate resources, coordination is another factor that positively moderates the dispersion-adaptation link. International firms should develop the competency to organize their dispersed capabilities and coordination mechanisms to exploit market needs effectively. Distributed valued capabilities across the firm boundaries increase the complexity of their management. Coordination is vital to guarantee that all functions within the organization embrace the dispersion issues in the firm's strategies and allows the creation, communication, and delivery of differentiated value in marketplaces (Krush et al., 2015). Thus, where marketers are considered as integrating agents that facilitate coordination by guiding the development and deployment of capabilities, marketing capabilities are integrating processes that facilitate coordination by speaking out actionable activities and producing organizational structures that rely on external parties to achieve firms' goals (Moorman & Day, 2016).

8. Managerial Implications

Management's task is to determine how best to develop and leverage firms' capabilities for competitive advantage (Moorman & Day, 2016). Compared with conventional marketing, instilling a relatively simple organization of IM capabilities may not pay off in delivering increased value. Instead, a more sophisticated structure in which capability-based IM activities are dispersed across firm boundaries may be more beneficial. Thus, there is a need for firms to focus attention on inter-organizational dispersion of IM capabilities achieved through connecting to third parties' resources. Such dispersion has impacts on various organizational dimensions such as marketing structure, strategies, and bottom-line outcomes.

By using dispersion, firms can access other parties' capabilities to better affect their performance. However, simply dispersing capabilities and expecting such dispersion to benefit outcomes may not necessarily breed performance success as dispersion affects

firms' sales and profits indirectly through firms' adaptation strategies. Top management roles need to be broader and consider circumstances how the firm IM activities should be (re)organized in order to achieve results in accordance with the creation of superior value outcomes. IM capabilities dispersion as a DC can serve as a mechanism of organizational capability building. Specifically, firms can seek to match capabilities with superior adaptation. While larger dispersion levels can bring benefits that directly enhance adaptation, managers need to realize that under higher levels of market dynamism or intra-firm coordination, the dispersion may better pay up. More turbulent customer environment or greater levels of cross-functional coordination may be preferable conditions to engage in higher levels of dispersed capabilities. Firms that meet these challenges will facilitate advantages in determining the level of adaptation, an important strategic action, that will translate into increased sales and profits. While there are ongoing arguments among managers regarding the optimal adaptation level of the firm's marketing efforts in foreign markets, IM capabilities dispersion can be a means to control the degree of adaptation.

In sum, for DCs to operate effectively, managers must evaluate the position of the firms' current IM capability assets, that is, their location and structure (Liu et al., 2013). IM managers, as boundary spanners, should foresee scenarios in which marketing activities should span across their organizations and lead the responsibility for being supporters and integrators of all the pieces of overall activities. Due to the essential role of capability dispersion, it is important that firms appoint IM personnel that can manage dispersed capabilities. It is critical for firms to conduct periodic reviews of their foreign environments stability when making decisions on their make-or-buy capability choices regarding cross-border markets. Managers should also evaluate their cross-functional interface level when deciding on how much IM capabilities dispersion to undertake.

9. Limitations and further research

Our research reveals limitations and insights, which suggest several potentially

avenues for future research. First, we limit the examination of dispersion to six capabilities known to be related to performance. Notably, DC theory proposes managing various types of capabilities, which complement each other (Najafi-Tavani et al., 2016). While the literature does not propose a comprehensive list of IM capabilities, their number and type can be expanded. Specifically, following the evolution of digitization of marketing activities, digital marketing capabilities could be included. In addition, we study external parties without distinguishing their type (e.g., service vs. goods providers) or their form of contact with the focal firm (e.g., contractual, joint venture). Conducting a study on certain types and forms of contact may assist in responding the question what type of providers should be involved in and responsible for executing certain IM activities.

Second, IM capabilities dispersion implies inter-firm relationships. While we investigate two certain moderating factors for the dispersion-adaptation link, other moderators such as organizational (e.g., task complexity, time to internationalization) or behavioral (e.g., trust, conflict, communication frequency) could be considered. Notably, our market dynamism moderator represents only one facet of market turbulence. While it is a major facet, other facets (e.g., technological) could be included. Similarly, our coordination moderator reflects intra-firm mechanisms. However, inter-firm coordination could be examined. Moderators of the direct dispersion-performance link could be also viewed. In addition, we designed adaption as a downstream effect of IM capabilities dispersion. Other strategic IM activities (e.g., market orientation) may be considered.

Third, our study focuses on inter-firm IM capabilities dispersion. Simultaneous examination of IM capabilities dispersion across both intra-firm (e.g., headquarter and international subsidiaries) and inter-firm boundaries could prove insightful since the form of dispersion matters and its affect may vary (Krush et al., 2015). Furthermore, this study examines dispersion in an IM context, which differs from a non-IM context. A

simultaneous consideration of IM and non-IM inter- and intra-firm may be essential.

Fourth, performance was measured subjectively by two clean common single indicators. While the use of single versus multi-item measures for performance is in debate among scholars (Katsikeas et al., 2000), employing multi-item scales and/or objective indicators tapping different aspects of performance can help future research examine performance trade-offs of IM strategy adaptation and dispersion (Katsikeas et al., 2016).

Fifth, our model's constructs may be affected by contextual cultural factors. National culture influences the way firms organize activities, shapes firms' orientations (e.g., adaptation), and affects how firms coordinate their activities (Schmid et al., 2016). While we focus on a certain country, our findings may need further validation in others. Hence, a culture sensitive study may extend and generalize our findings. Moreover, cross-national/cultural analyses could be also carried out to examine if our results are challenged.

Finally, we use the DC view as a conceptual framework. Dispersion can be tackled from additional perspectives. For example, it can be viewed as a form of networking and thus a network approach can be undertaken. Moreover, the globalization of markets and firms' increased reliance on outsourcing to gain competitive advantage make companies vulnerable to unknown risks and their impact. For example, as firms involve external partners in their activities, they expose their strategies to imitation. In addition, dispersing marketing functions carries obvious drawbacks (e.g., costs of operating abroad). Our study does not consider such risks and drawbacks. Future research can adopt a risk management perspective and consider disadvantages to investigate IM capabilities dispersion. As a related aspect, dispersion raises concerns regarding the control (e.g., proprietary of assets) and power issues between the focal firm and its partners (Pitelis & Teece, 2017; 2018). While these are important topics in international studies, our IM capabilities dispersion construct did not explicitly encompass these aspects, and future research could.

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Table 1 - Descriptive, correlation, and statistic data

Construct	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Firm size	466	827	1.00												
2. Firm age	38	19	.19**	1.00											
3. IM firm experience	28	13	.19 *	.41**	1.00										
4. Differentiation strategy	n.a.		.05	-.03	-.10	1.00									
5. Cost leadership strategy	n.a.		-.04	.08	.10	-.42*	1.00								
6. B2B/B2C	2.28	0.5	-.03	.21	-.12	-.09	-.07	1.00							
7. Industry	n.a.		.09	-.06	.07	.03	.06	.13	1.00						
8. IM Capably Dispersion	4.55	1.2	.08	.09	-.04	.24	-.18	-.10	.01	1.00					
9. Adaptation	4.62	1.3	-.04	-.01	.01	.29*	-.20*	-.26**	.03	.33**	1.00				
10. Market Dynamism	4.68	0.9	-.03	.03	.03	.05	.11	.06*	.05	.39**	.43**	1.00			
11. IM Coordination	4.87	1.1	.16	-.04	.16*	.09	.04	.04	.09	.18*	.21*	.03	1.00		
12. Int. sales performance	5.55	1.2	.05	-.04	-.10	.22	-.03	-.21**	.06	.38**	.36**	.26**	.39*	1.00	
13. Int. profit performance	4.96	0.8	-.08	-.14	-.04	.17	-.06	-.17**	.03	.29**	.31**	.19**	.25*	.38**	1.00
Cronbach alpha			n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	.89	.95	.86	.85	n.a.	n.a.
Composite reliability			n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	.88	.83	.74	.80	n.a.	n.a.
Average variance extracted (AVE)			n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	.61	.58	.57	.59	n.a.	n.a.
Square root AVE										.78	.76	.75	.77		

** Significant at the 0.01 level

* Significant at the 0.05 level

n.a. - Not applicable for a single item measure

Table 2 - Effects on adaptation and international performance

	Adaptation			Performance					
				Sales			Profits		
				Controls	With IM Dispersion Only	With IM Dispersion and Adaptation	Controls	With IM Dispersion Only	With IM Dispersion and Adaptation
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Control Variables									
Firm size	-.047	-.049	-.051	.056	.051	.049	-.048	-.051	-.061
Firm age	.012	-.009	.006	-.009	-.009	-.005	-.070	-.067	-.071
IM firm experience	.013	.015	.009	-.086	-.079	-.080	-.028	-.027	-.021
Differentiation strategy	.106**	.128*	.089	.095*	.061*	.060	.188*	.171*	.129
Cost leadership strategy	-.090*	-.068*	-.055	-.055*	.062	.056	-.080*	.091	.089
B2B/B2C	-.128**	-.121*	-.120*	-.099*	-.091*	-.111	-.111*	-.107	-.100
Industry	.069	.065	.060	.061	.058	.057	-.018	-.019	-.021
Market Dynamism X Adaptation				.251**	.161**	.199*	.261**	.209**	.128*
IM Coordination X Adaptation				.151*	.181*	.187*	.069*	.091*	.076*
Main Effects									
IM Capabilities Dispersion		.210**	.143*		.187**	.181		.119*	.124
Market Dynamism		.264*	.278*						
IM Coordination		.070	.063						
Adaptation						.137**			.099**
Moderation Effects									
IM Dispersion X Dynamism			.601**						
IM Dispersion X Coordination			.134*						
R ²	.127	.210	.378	.272	.298	.321	.299	.315	.333
R ² Adjusted	.100	.172	.342	.235	.256	.277	.269	.279	.296
Δ R ²		.083	.168		.026	.023		.016	.018
Sig. of Δ R ²		p ≤ .0	p ≤ .0		p ≤ .0	p ≤ .0		p ≤ .0	p ≤ .0

* p ≤ .05, ** p ≤ .01, one-tailed test was used because the hypotheses are directional

Appendix - Construct measurement

Inter-Organizational IM Capabilities Dispersion (Krush et al., 2015) <i>To what extent do overseas independent organizations such as consultants, agencies, providers and other firms contribute to the firm's (1-No extent, 7-High extent):</i> IM information management capability International channel management capability IM communication capability International product development capability International pricing capability IM planning capability
Marketing Program Adaptation (Gnizy et al., 2014) <i>To what extent your firm usually adapt each of the following elements of the marketing program in its most important international market(s) (1-No adaptation at all, 7-Extensive adaptation):</i> Size of product/service line Product/process design Product positioning Brand name and/or packaging Price Advertising and/or sales promotion Sales force structure and management Upstream supply chain (parts/process suppliers) Downstream supply chain (wholesalers, retailers) Customer service
Market Dynamism (Spyropoulou et al., 2018) <i>In our international markets (1 = Not at all; 7 = To an extreme extent):</i> Customer needs and demands are changing rapidly. New markets are emerging for products and services.
IM Coordination (Cadogan et al., 2012) <i>Consider the interaction between functional areas/departments in your firm (1 = Not at all; 7 = To an extreme extent):</i> IM employees and those in other functional areas help each other out. With an emphasis on IM operations, in this company there is a sense of teamwork going right down to the 'shop floor'. There is a strong collaborative working relationship between IM and other functional areas. Functional areas in this company pull together in the same direction in IM operations. The activities of our business functions are integrated in pursuing common goals in IM operations. We resolve issues and conflicts regarding IM operations through communication and group problem solving.
International Sales Performance <i>How would you rate your firm:</i> Average annual international sales growth compared to the industry average (1-Poor, 7-Outstanding)
International Profit Performance Overall, how profitable has international operation been during the last financial year (1-Very unprofitable, 7-Very profitable)
Controls (single-item scales) Firm size* - number of employees Firm age* - number of years IM firm experience* - number of years a firm had been involved in IM Business strategy (adapted from Krush et al., 2015) - what is the generic strategy most applicable to your firm: <i>Cost leadership</i> (obtaining the lowest costs in the market) or <i>Differentiation</i> (focusing on being better in different features of the offering that are important to customers) B2B/B2C - Please rate your firm focus: 1- turnover totally from B2B, 7- turnover totally from B2C Industry - a dummy calculated variable: hi-tech or not

* Log-transformed due to the construct's large mean and SD (Table 1) relative to the Likert-scale variables, as well as its non-normal and truncated distribution

Figure 1 - Research model

